

1. Dataset Shape

Medical_Appointment_NoShow.ipynb

Colab output

```
[9] import pandas as pd

df = pd.read_csv("Medical_appointment_data.csv")

print(df.shape)
df.head()
```

(109593, 26)

	specialty	appointment_time	gender	no_show	disability	\
0	psychotherapy	17	F	yes	intellectual	
1	NaN	7	M	no	intellectual	
2	speech therapy	16	M	no	intellectual	
3	speech therapy	14	M	yes	intellectual	
4	physiotherapy	8	M	no	motor	

	place	appointment_shift	age	under_12_years_old	\
0	Lake Marvinville	afternoon	9.0	1	
1	ITAPEMA	morning	11.0	1	
2	ITAJAÍ	afternoon	8.0	1	
3	Sarahside	afternoon	9.0	1	
4	ITAJAÍ	morning	NaN	0	

	over_60_years_old	...	storm_day_before	rain_intensity	heat_intensity	\
0	0	...	1	no_rain	warm	
1	0	...	1	no_rain	cold	
2	0	...	1	no_rain	warm	
3	0	...	1	moderate	mild	
4	0	...	1	no_rain	mild	

	appointment_date_continuous	Hipertension	Diabetes	Alcoholism	Handcap	\
0	2020-01-01	0	0	0	0	
1	2020-01-01	0	0	0	0	
2	2020-01-01	0	0	0	0	
3	2020-01-01	0	0	0	0	
4	2020-01-01	0	0	0	0	

	Scholarship	SMS_received
0	0	0
1	0	0
2	0	0
3	0	1
4	0	0

[5 rows x 26 columns]

2. Missing Values Handling

Medical_Appointment_NoShow.ipynb

Colab output

```
[11] df.isnull().sum()
```

```
specialty          20127
appointment_time    0
gender              0
no_show            0
disability          16601
place              11539
appointment_shift   0
age                22960
under_12_years_old  0
over_60_years_old  0
patient_needs_companion 0
average_temp_day    2211
average_rain_day    2245
max_temp_day        2227
max_rain_day         2263
rainy_day_before    0
storm_day_before    0
rain_intensity       0
heat_intensity       0
appointment_date_continuous 0
Hipertension         0
Diabetes              0
Alcoholism            0
Handcap              0
Scholarship           0
SMS_received         0
dtype: int64
```

```
[29] for col in df.columns:

    if df[col].dtype == "object":
        df[col] = df[col].fillna(df[col].mode()[0])

    else:
        df[col] = df[col].fillna(df[col].median())

print("Missing values handled")
```

```
Missing values handled
```

3. Model Training

Medical_Appointment_NoShow.ipynb

Colab output

```
[32] from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("Train:", X_train.shape)
print("Test:", X_test.shape)
```

Train: (87674, 25)
Test: (21919, 25)

```
[33] from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

model.fit(X_train, y_train)

print("Model Trained Successfully")
```

Model Trained Successfully

4. Accuracy & F1 Score

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Colab output

```
[34] from sklearn.metrics import accuracy_score, f1_score

pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, pred))
print("F1 Score:", f1_score(y_test, pred))
```

```
Accuracy: 0.7153611022400657
F1 Score: 0.47229975471538527
```

5. Forecast Model Results

Medical_Appointment_NoShow.ipynb

Colab output

```
[38] daily = df.groupby("appointment_date_continuous").size().reset_index()

daily.columns = ["date", "appointments"]

daily.head()
```

	date	appointments
0	0	25
1	1	33
2	2	18
3	3	16
4	4	15

```
[44] from sklearn.ensemble import RandomForestRegressor

forecast_model = RandomForestRegressor(
    n_estimators=100,
    random_state=42
)

forecast_model.fit(X_train, y_train)

print("Forecast Model Trained")
```

Forecast Model Trained

```
[46] joblib.dump(
    forecast_model,
    "demand_forecast_model.pkl"
)

print("Demand Forecast Model Saved")
```

Demand Forecast Model Saved